

Arunabha Majumder

Mechatronics & Robotics Engineer | Mechanical Design | Actuators & Control | Prototyping

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

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Application: Mechatronics & Robotics Engineer

Dear Hiring Manager,

I am a mechatronics and robotics engineer with a BTech in Mechanical Engineering and an MTech in Mechatronics, and I am finishing my PhD in Mechatronics and Robotics at Aalborg University. For more than five years I have worked hands-on with robotic and electromechanical systems, taking ideas from concept to tested hardware, and I am now looking to bring that experience to a robotics or mechatronics company in Denmark.

In my PhD, I designed a hybrid-actuated shoulder exoskeleton that combines a parallel spring with a small 6 Nm motor. I created the mechanical design in SolidWorks, used FEA to iterate for stiffness, strength and weight, built the prototype from machined and 3D-printed parts, and integrated the motor, encoder, force sensors, DAQ and embedded control. The design cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also implemented an Assist-as-Needed controller that reduced user muscle effort by more than 15%.

I enjoy working where mechanics, sensing and control meet. On a 5-bar planar parallel robot, I estimated the end-effector force from variable-stiffness actuators through the robot's Jacobian, and validated the estimate against a force sensor to within 10% RMS error. During my Master's at CSIR-CMERI, I fabricated pneumatic artificial muscles and developed a neural-network-based controller that reduced tracking error to below 0.8%.

I work in a structured way, from test planning to data analysis, and I communicate clearly with colleagues from different disciplines. I have planned and run a 12-participant validation study, taught and supervised Bachelor's students, and published peer-reviewed work. I am a Certified SolidWorks Associate and fluent in English.

I would welcome the opportunity to discuss how I can contribute to your team. My portfolio and project website are linked above, and references are available upon request.

Kind regards,

Arunabha Majumder